

Machine Learning Model Outcomes

Executive summary report for the New York City Taxi and Limousine Commission
Prepared by Automatidata

Overview

Automatidata evaluated and built a machine learning model for the New York City Taxi & Limousine Commission (NYC TLC) to predict which credit card passengers will leave a generous tip ($\geq 20\%$).

Problem

Drivers depend on tips, but predicting tipping behavior is hard. Predicting non-tippers creates serious risks, such as drivers refusing rides to certain passengers and flawed predictions caused by missing cash tip data

Solution

We shifted the objective to predict generous tippers using credit card trips only. We trained Random Forest and XGBoost models on pre-trip information so drivers can see positive earnings opportunities without causing service discrimination.

Details

Behind the data

- Filtered data to include credit card payments only, as cash tips are recorded as \$0.
- Engineered pre-trip features including pickup/dropoff locations, time of day, estimated trip distance, and estimated fare.
- Created a binary target variable **generous** (1 for tips $\geq 20\%$, 0 for tips $< 20\%$).

	model	precision	recall	F1	accuracy
0	RF	0.658636	0.855012	0.744026	0.690221
0	RF_test	0.628378	0.868077	0.729031	0.660334
0	XGB	0.692464	0.800403	0.742458	0.707745
0	XGB test	0.685518	0.807094	0.741355	0.703570

Image Alt-Text: F1 scores for random forest and XGboost models

Future model suggestions

- Collect external features like real-time traffic delays, weather conditions, and passenger trip history.

Results Summary

Champion Model: XGBoost was chosen over Random Forest due to higher precision and overall accuracy.

Error Analysis: The model makes more false positive errors (predicting a high tip when the rider tips less) than false negative errors.

Next Steps

It is a starting tool, not a final production model. Adjust the classification decision threshold (raise it above 0.5) to make the model stricter before predicting a high tip, or tune hyperparameters specifically to optimize for precision.